

DS	Session	Searching	Question Information	Level 1	Challenge 4
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Question Description:

Ragu has given a range [L, R] to Smith. Smith wants to require to find the number of integers 'X' in the range such that $GCD(X, F(X)) > 1$ where F(x) is equal to the sum of digits of 'X' in its hexadecimal (or base 16) representation.

Example : $F(27) = 1+B=1+11=12$

(27 in hexadecimal is written as 1B)

Constraints:

$$1 \leq T \leq 50$$

$$1 \leq L$$

$$R \leq 10^5$$

Input Format:

The first line contains a positive integer 'T' denoting the number of questions that you are asked.

Each of the next 'T' lines contain two integers 'L' and 'R' denoting the range of questions.

Output Format:

Print the output in a separate lines exactly 'T' numbers as the output.

Sample Input:

3
1 3
5 8
7 12

Sample output:

2
4
6

Explanation

For the first test-case, numbers which satisfy the criteria specified are : 2,3. Hence, the answer is 2.

For the second test-case, numbers which satisfy the criteria specified are : 5,6,7 and 8. Hence, the answer is 4.

For the third test-case, numbers which satisfy the criteria specified are : 7,8,9,10,11 and 12. Hence, the answer is 6.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 3
5 8
7 12

Test Case 2

INPUT (STDIN)

10
15 64
14 100
9 79
19 63
3 83
5 61
18 93

EXPECTED OUTPUT

2
4
6

20 98
13 71
12 53

EXPECTED OUTPUT

29
52
45
26
53
37
45
47
36
26

✓ Mandatory Test Cases

Test Case 1

KEYWORD

`int search(int a, int b)`

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

165

Test Case 3

NLOC

27

Searching4.c +

```
1 #include <stdio.h>
2
3 int gcd(int a, int b) {
4     while (b) {
5         int t = a % b;
6         a = b;
7         b = t;
8     }
9     return a;
10}
11
12int f_hex(int x) {
13    int sum = 0;
14    while (x > 0) {
15        sum += (x % 16);
16        x /= 16;
17    }
18    return sum;
19}
20
```

main | CH.SC.U4CSE25225 | Searching4.c

Ln 40, Col 1 Spaces: 4 UTF-8 CRLF C

```
22 int count = 0;
23 for (int x = a; x <= b; x++) {
24     if (gcd(x, f_hex(x)) > 1) {
25         count++;
26     }
27 }
```

```
26     }
27 }
28 return count;
29}
30
31int main() {
32     int t, l, r;
33     if (scanf("%d", &t) != 1) return 0;
```

```
34 while (t--) {  
35     scanf("%d %d", &l, &r);  
36     printf("%d\n", search(l, r));  
37 }  
38 return 0;  
39}  
40
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

1 3

5 8

7 12

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching4.exe

2

4

6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

10

15 64

14 100

9 79

19 63

3 83

5 61

18 93

20 98

13 71

12 53

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching4.exe

29

52

45

26

53

37

45

47

36

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```